Network-Channel Privacy Leakage — Stage Gaps and Invariant Results

Status: Final — 144,396 attr-item pairs across 6,876 unique items, 24 apps, 10 datasets. Generated: 2026-05-06. Analysis scoped exclusively to the **NETWORK channel**.

1. Experimental Setup

The Lantern evaluation pipeline captures app behavior across four externalization channels: **NETWORK**, **UI**, **STORAGE**, and **LOGGING.** This analysis focuses exclusively on the **NETWORK** channel — outbound HTTP/API calls carrying private attribute signals.

Metric	Value
Attr-item pairs evaluated	**144,396**
Unique items	**6,876**
Apps covered	**24**
Datasets	**10**
NETWORK-present attr-item pairs	**87,066** (60% of all)
Unique items with NETWORK captured	**4,146**
Apps with ≥1 NETWORK item	**18**
Apps with zero NETWORK presence	**6** (clone, google-ai-edge-gallery, momentag, pocketpal-ai, skin-disease-detection, snapdo)

Stage definitions.

1. **Raw output** — attribute signal present in the LLM's direct textual response (`output_eval`).
2. **Any-channel confirmed** — attribute confirmed leaked on at least one captured channel (`ext_conf`).
3. **NETWORK confirmed | present** — attribute confirmed leaked specifically on the NETWORK channel, conditioned on the channel being present.

All leakage rates are computed over (item, attribute) pairs. Channel-level rates are **conditioned on the channel being present** (i.e., the app made at least one network call in that run) to avoid dilution from structurally absent channels.

2. Headline Numbers

Metric	Rate
Raw output any-leak (all pairs)	**18.9%**
Any-channel confirmed (all pairs)	**4.2%**
NETWORK confirmed | present	**0.79%**
NETWORK any-leak | present	**2.66%**

Minimum raw→NETWORK compression	**7.9× (Health)**

The NETWORK channel is the most disciplined externalization point: models infer private attributes at 18.9% in raw output, but only 0.79% of attribute-item pairs on the NETWORK channel receive a confirmed-leakage verdict — a compression of at least **7.9× in every single app category**.

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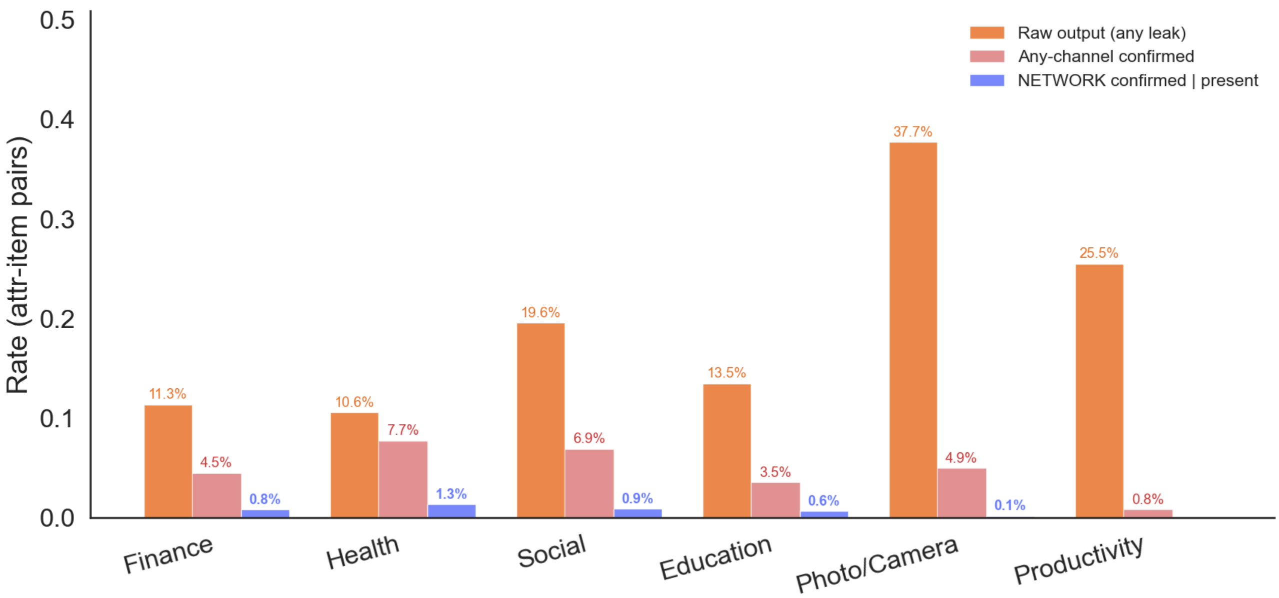
3. Key Findings

Finding 1 — The raw→NETWORK stage gap is an invariant across all categories

In every app category, the raw-output leak rate exceeds the NETWORK confirmed rate by at least 7.9×. The gap ranges from 7.9× (Health) to effectively infinite (Productivity: 25.5% raw → 0.0% NETWORK). This means the NETWORK channel is **not** a simple pass-through of the model's inference.

Category	N pairs	Raw output leak	Any-ext conf	NETWORK conf|pres	Compression
Finance	31,731	11.3%	4.5%	0.81%	14×
Health	16,800	10.6%	7.7%	1.33%	8×
Social	16,443	19.6%	6.9%	0.86%	23×
Education	31,710	13.5%	3.5%	0.64%	21×
Photo/Camera	18,438	37.7%	4.9%	0.15%	251×
Productivity	29,274	25.5%	0.8%	0.00%	∞

Implication: The NETWORK channel exerts meaningful selective pressure on what attributes exit the app. However, the residual NETWORK leakage (0.79% confirmed) is non-zero and affects real sensitive attributes — so it cannot be ignored.



Finding 2 — Identity dominates NETWORK in conversational apps; domain context shifts rankings in Health

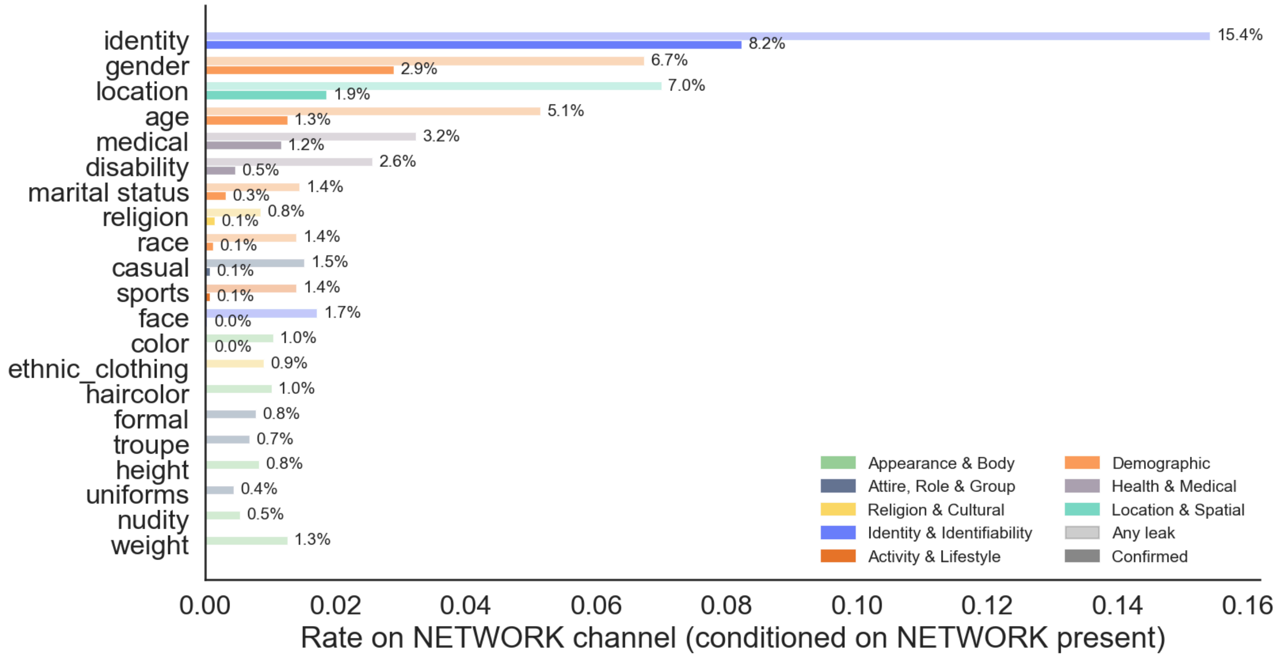
Pooled across all NETWORK-present pairs, `identity` leads at **8.2% confirmed** — 2.8× the next-highest attribute (`gender` at 2.9%). However this aggregate masks a domain-specific split: in Finance, Education, and Social categories `identity` accounts for **54–70% of all confirmed NETWORK pairs**, while in Health apps the top three are `gender`, `age`, and `medical` — reflecting that health apps process patient consultation context in which demographic and clinical attributes dominate.

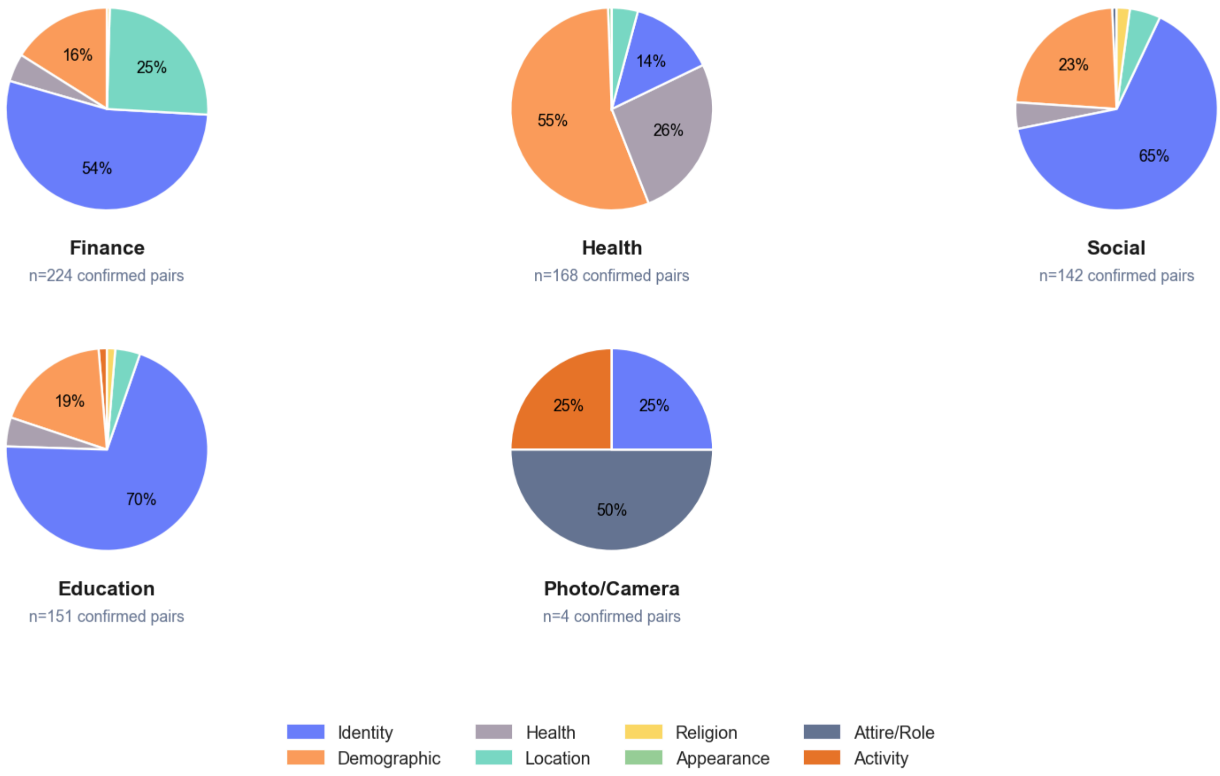
Category	Top NETWORK attr	Confirmed rate	Identity share of NETWORK confirmed
Finance	`identity`	9.1%	54%
Health	`gender`	7.5%	14%
Social	`identity`	11.7%	65%
Education	`identity`	9.5%	70%
Photo/Camera	`casual`	1.6%	0%
Productivity	`age`	0.0%	—

Full attribute breakdown (pooled over all NETWORK-present pairs):

Attribute	Family	N (NETWORK present)	Any leak	Confirmed
`identity`	Identity & Identifiability	4,146	15.41%	8.22%
`gender`	Demographic	4,146	6.73%	2.89%
`location`	Location & Spatial	4,146	6.99%	1.86%
`age`	Demographic	4,146	5.14%	1.25%
`medical`	Health & Medical	4,146	3.23%	1.16%
`disability`	Health & Medical	4,146	2.56%	0.46%
`marital status`	Demographic	4,146	1.45%	0.31%
`religion`	Religion & Cultural	4,146	0.84%	0.14%
`race`	Demographic	4,146	1.40%	0.12%
`casual`	Attire, Role & Group	4,146	1.52%	0.07%
`sports`	Activity & Lifestyle	4,146	1.40%	0.07%
`face`	Identity & Identifiability	4,146	1.71%	0.02%
`color`	Appearance & Body	4,146	1.04%	0.02%
`nudity`	Appearance & Body	4,146	0.53%	0.00%
`ethnic_clothing`	Religion & Cultural	4,146	0.89%	0.00%
`haircolor`	Appearance & Body	4,146	1.01%	0.00%
`formal`	Attire, Role & Group	4,146	0.77%	0.00%
`troupe`	Attire, Role & Group	4,146	0.68%	0.00%
`height`	Appearance & Body	4,146	0.82%	0.00%
`uniforms`	Attire, Role & Group	4,146	0.43%	0.00%
`weight`	Appearance & Body	4,146	1.25%	0.00%

Implication: NETWORK-channel defenses in conversational apps should prioritize identity redaction. Health apps need a different profile: demographic and clinical attribute filters are more urgent. In all categories, appearance and attire attributes show zero or near-zero NETWORK confirmed leakage and can be deprioritized.





Finding 3 — Two distinct mechanisms: persistence vs inference injection

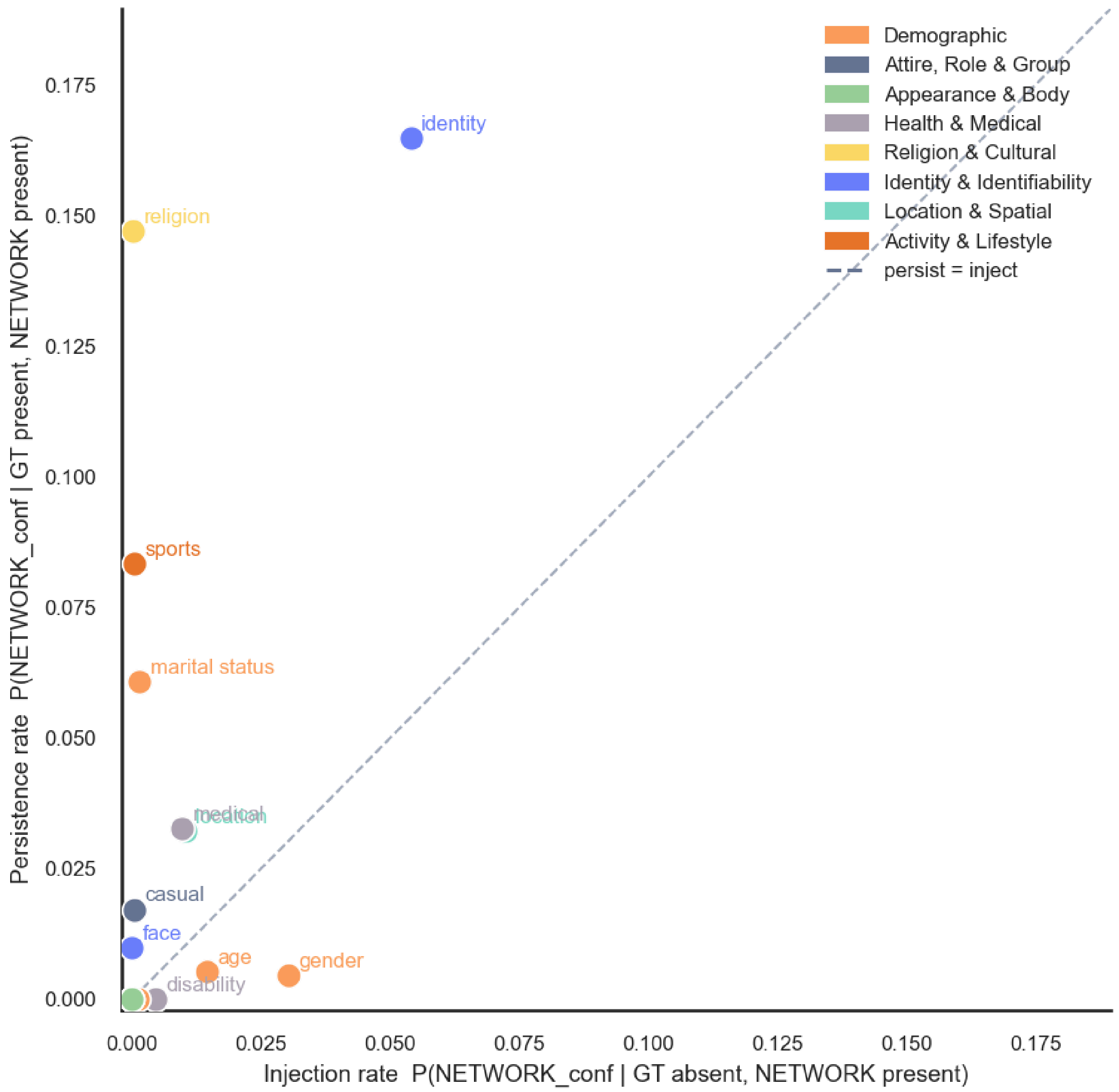
Attributes cluster into two mechanistic regimes:

- **Persist-dominant** (`identity`, `religion`, `marital status`): the attribute reaches the NETWORK channel primarily when it was already present in the input GT. Persistence rates exceed injection rates by 2–8×.

- **Inject-dominant** (`gender`, `age`, `disability`): the attribute appears on the NETWORK channel even when the input GT does NOT contain it. Gender's injection rate (3.0%) is 7× its persistence rate (0.5%), meaning NETWORK gender leakage is **primarily model inference, not data pass-through**.

Attribute	Family	Persistence	Injection	Regime
`identity`	Identity & Identifiability	16.49%	5.40%	Both high
`religion`	Religion & Cultural	14.71%	0.02%	Persist-dominant
`sports`	Activity & Lifestyle	8.33%	0.05%	Persist-dominant
`marital status`	Demographic	6.09%	0.15%	Persist-dominant
`medical`	Health & Medical	3.28%	0.97%	Persist-dominant
`location`	Location & Spatial	3.23%	1.04%	Persist-dominant
`casual`	Attire, Role & Group	1.72%	0.05%	Persist-dominant
`face`	Identity & Identifiability	0.99%	0.00%	Persist-dominant
`age`	Demographic	0.54%	1.46%	Inject-dominant
`gender`	Demographic	0.46%	3.03%	Inject-dominant
`disability`	Health & Medical	0.00%	0.46%	Inject-dominant

Implication: Filtering strategies must differ by regime. Persist-dominant attributes require redaction at the data-ingestion layer (before model processing). Inject-dominant attributes require output-side filtering, because the model spontaneously generates them regardless of input content.



Finding 4 — NETWORK presence is structurally binary across apps

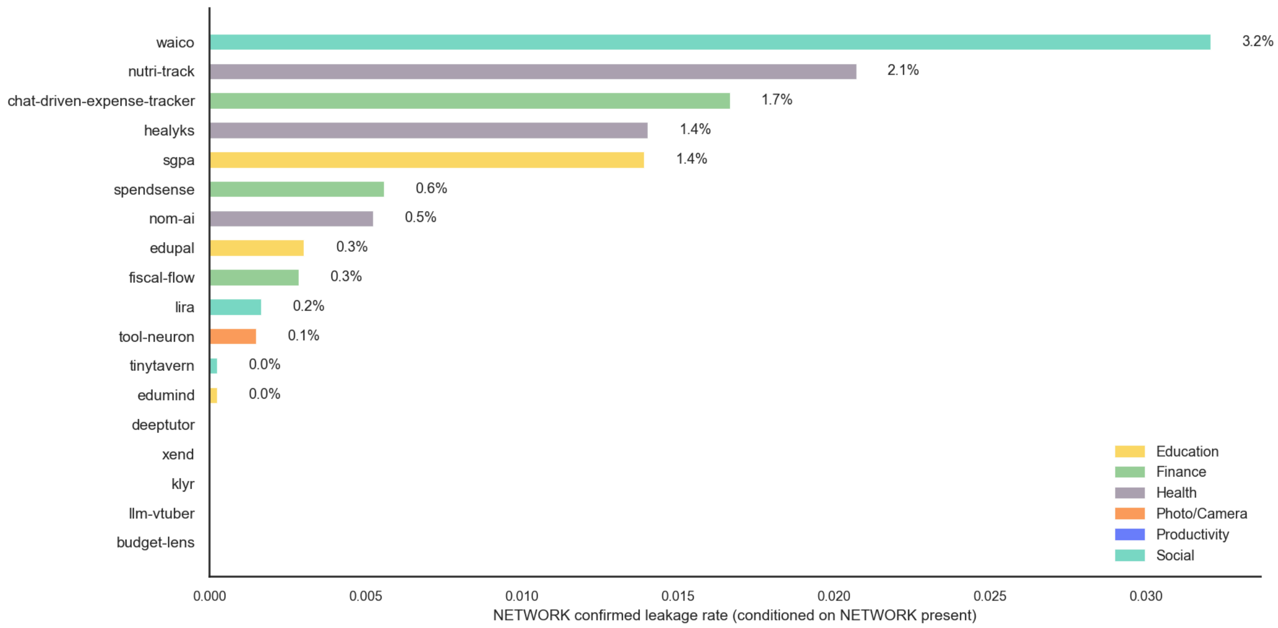
Apps fall into two sharply separated groups: those that route every request through a cloud API (**100% NETWORK presence**: 14 apps) and those that run fully on-device with **zero NETWORK presence** (6 apps: clone, google-ai-edge-gallery, momentag, pocketpal-ai, skin-disease-detection, snapdo). Only `budget-lens` (6.5%) and `xend` (1.0%) occupy intermediate ground, where NETWORK calls occur sporadically.

Among apps with NETWORK presence, confirmed leakage ranges from 0.00% (`klyr`, `llm-vtuber`) to 3.21% (`waico`).

App	Category	NETWORK items	Raw output	NETWORK conf|pres
waico	Social	199	16.7%	3.21%
nutri-track	Health	200	20.8%	2.07%
chat-driven-expense-tracker	Finance	400	20.5%	1.67%

healyks	Health	200	10.9%	1.40%
sgpa	Education	400	12.3%	1.39%
spendsense	Finance	511	6.6%	0.56%
nom-ai	Health	200	0.1%	0.52%
edupal	Education	518	13.7%	0.30%
fiscal-flow	Finance	400	9.7%	0.29%
lira	Social	200	24.1%	0.17%
tool-neuron	Photo/Camera	127	45.7%	0.15%
tinytavern	Social	194	19.5%	0.02%
edumind	Education	197	17.4%	0.02%
llm-vtuber	Social	190	17.8%	0.00%
deeptutor	Education	1	12.3%	0.00%
xend	Productivity	2	24.7%	0.00%
klyr	Productivity	194	0.4%	0.00%
budget-lens	Finance	13	8.5%	0.00%

Implication: Whether an app poses NETWORK privacy risk is largely determined by its architecture (cloud vs. on-device). Within cloud-API apps, the top five by NETWORK confirmed rate (`waico`, `nutri-track`, `chat-driven-expense-tracker`, `healyks`, `sgpa`) account for the bulk of NETWORK leakage.



Finding 5 — NETWORK has higher confirmed rate than STORAGE and LOGGING, lower than UI

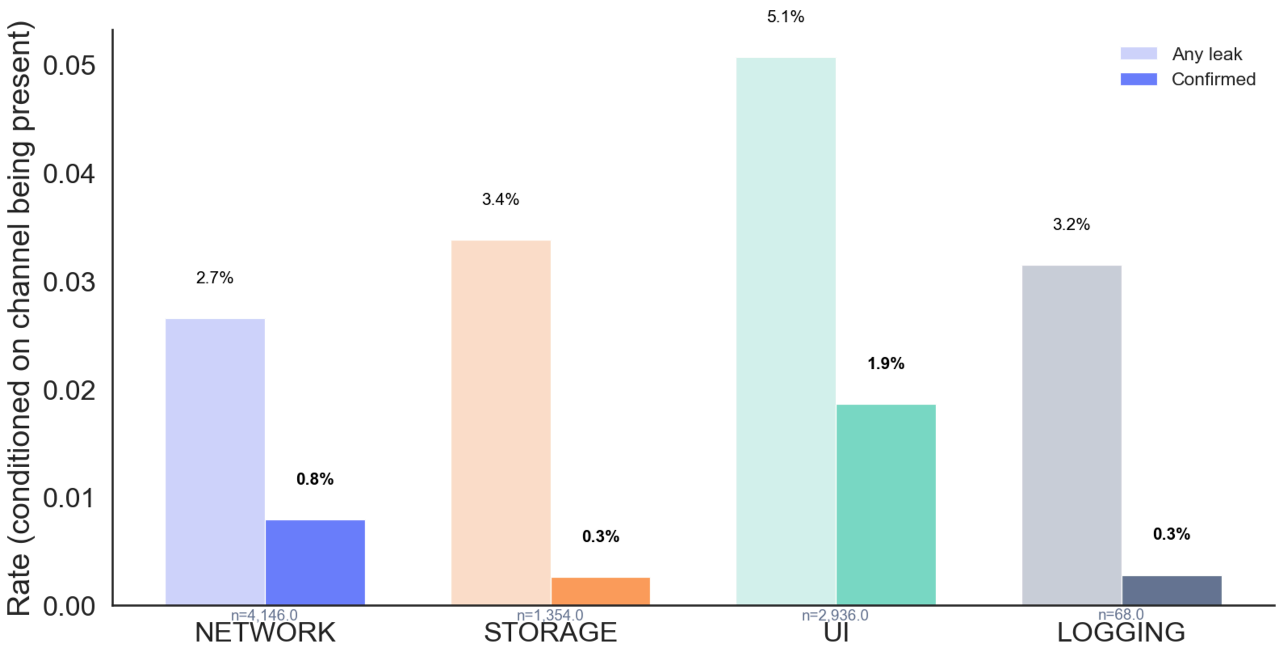
Among the four channels (conditioned on presence):

Channel	N pairs (present)	N items	Any leak	Confirmed
NETWORK	87,066.0	4,146.0	2.66%	0.79%

STORAGE	28,434.0	1,354.0	3.38%	0.26%
UI	61,656.0	2,936.0	5.07%	1.86%
LOGGING	1,428.0	68.0	3.15%	0.28%

NETWORK's confirmed rate (0.79%) exceeds STORAGE (0.26%) and LOGGING (0.28%) but falls below UI (1.86%). This ordering is invariant across categories: UI carries the most confirmed leakage because it renders the model's full output; NETWORK is selective (only API call payloads are captured); STORAGE and LOGGING are even more selective.

Implication: UI-channel filtering is the highest-priority intervention, but NETWORK confirmed leakage is structurally higher than STORAGE and LOGGING — making it the second-priority channel for mitigation.



Finding 6 — The NETWORK boundary suppresses visual and attire attributes completely; abstract semantic attributes survive

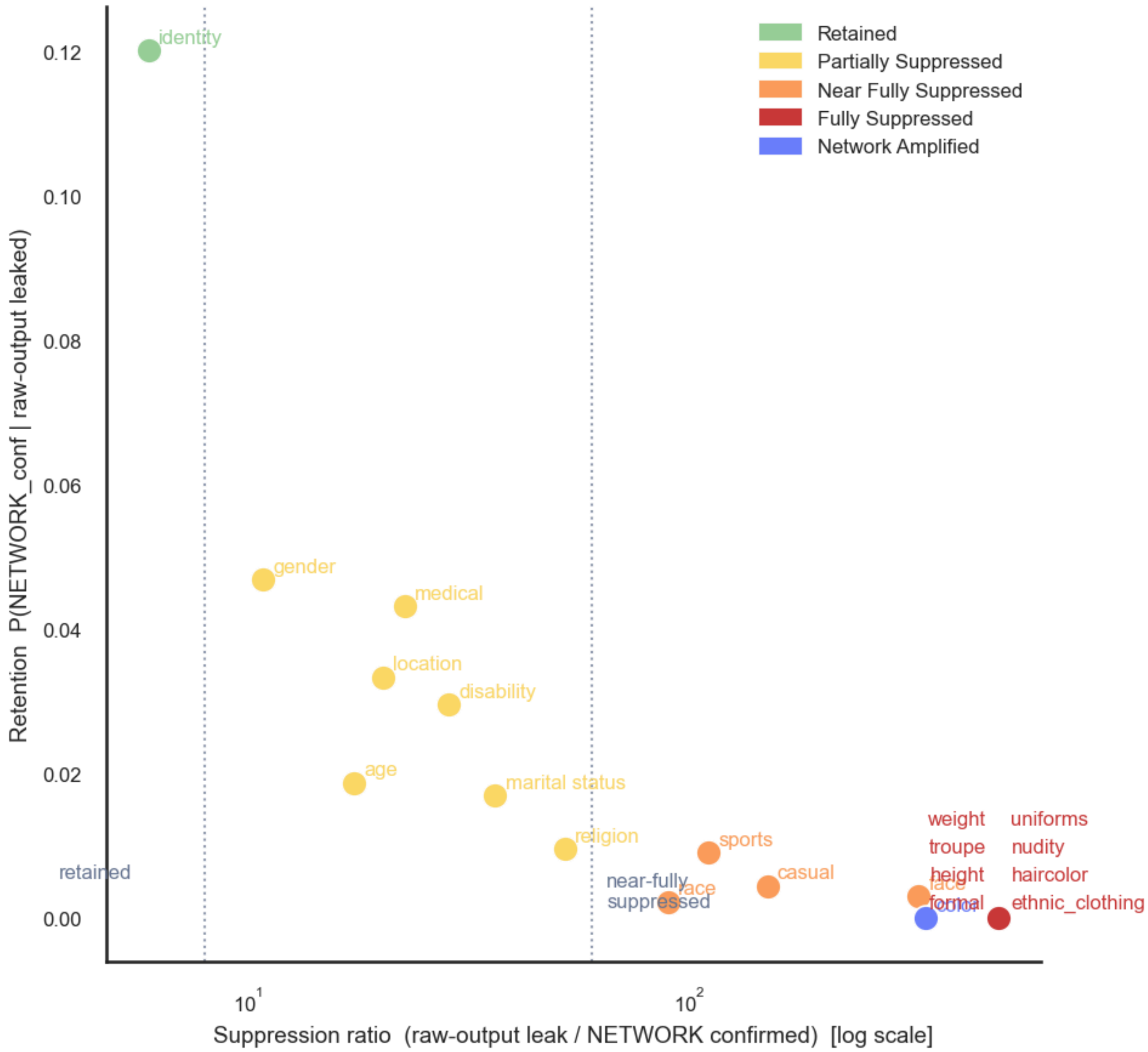
The NETWORK channel acts as a differential filter whose selectivity depends on attribute type. Classifying each attribute by its raw-output → NETWORK suppression ratio reveals four distinct regimes:

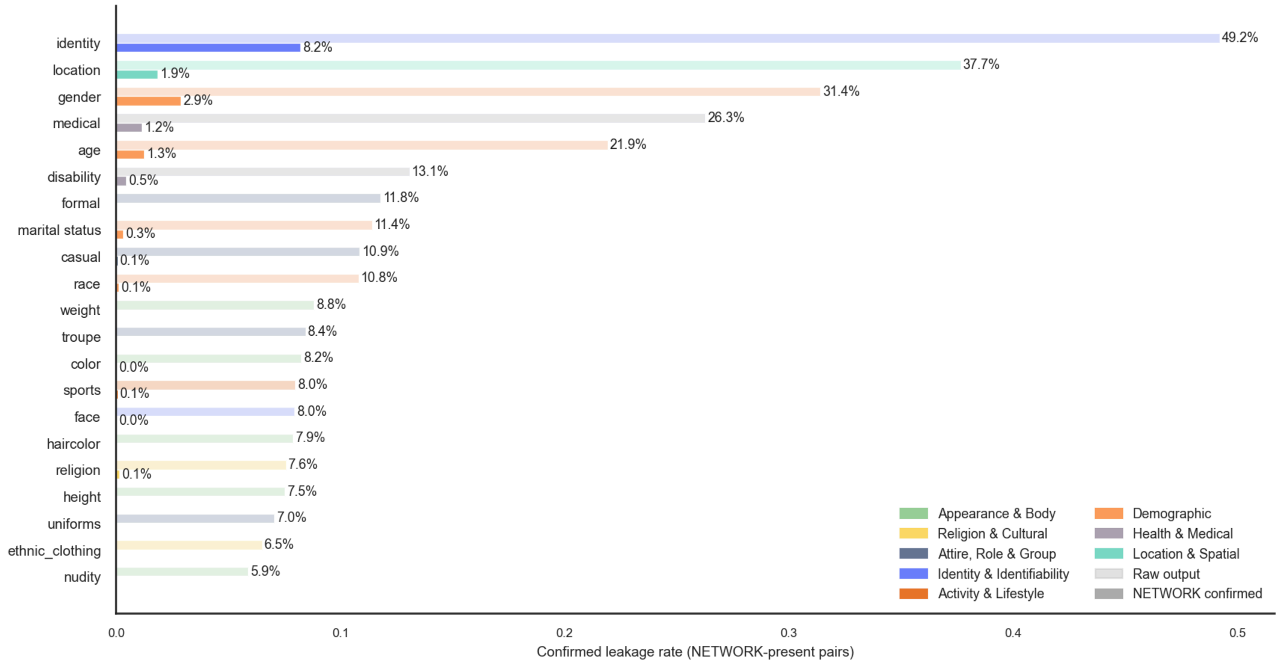
- **Retained** (suppression ratio < 8×): `identity`. These attributes survive the raw→NETWORK transition at the highest rate (12.0% average retention), meaning roughly 1 in 8 raw-output attribute signals makes it to a confirmed NETWORK verdict.
- **Partially suppressed** (8–60×): `gender`, `age`, `location`, `medical`, `disability`, `marital status`, `religion`. Sensitive semantic attributes with moderate filtering — they exit NETWORK but at heavily reduced rates. Average retention 2.8%.
- **Near-fully suppressed** (60–400×): `race`, `sports`, `casual`, `face`. Attributes that are present in raw output but almost entirely absent from NETWORK payloads. Appear in API calls only rarely.
- **Fully suppressed** (∞): `formal`, `haircolor`, `height`, `uniforms`, `ethnic_clothing`, `nudity`, `troupe`, `weight`. Zero confirmed NETWORK leakage despite raw-output rates of 6–12%. These are all physical/appearance/attire attributes that are verbally described by the model but stripped from — or never included in — API call payloads.

- **Network-amplified** (injection > retention): `color`. These attributes appear confirmed on NETWORK at higher rates than their raw-output signal alone would predict — the NETWORK payload adds or surfaces additional attribute information.

Attribute	Family	Raw conf	NETWORK conf	Suppression	Retention	New inject	Regime
`identity`	Identity & Identifiability	49.2%	8.22%	6×	12.0%	4.56%	retained
`gender`	Demographic	31.4%	2.89%	11×	4.7%	2.07%	partially suppressed
`age`	Demographic	21.9%	1.25%	17×	1.9%	1.08%	partially suppressed
`location`	Location & Spatial	37.7%	1.86%	20×	3.3%	0.97%	partially suppressed
`medical`	Health & Medical	26.3%	1.16%	23×	4.3%	0.03%	partially suppressed
`disability`	Health & Medical	13.1%	0.46%	29×	3.0%	0.08%	partially suppressed
`marital status`	Demographic	11.4%	0.31%	36×	1.7%	0.14%	partially suppressed
`religion`	Religion & Cultural	7.6%	0.14%	52×	1.0%	0.08%	partially suppressed
`race`	Demographic	10.8%	0.12%	90×	0.2%	0.11%	near-fully suppressed
`sports`	Activity & Lifestyle	8.0%	0.07%	110×	0.9%	0.00%	near-fully suppressed
`casual`	Attire, Role & Group	10.9%	0.07%	150×	0.4%	0.03%	near-fully suppressed
`face`	Identity & Identifiability	8.0%	0.02%	330×	0.3%	0.00%	near-fully suppressed
`color`	Appearance & Body	8.2%	0.02%	342×	0.0%	0.03%	network-amplified
`formal`	Attire, Role & Group	11.8%	0.00%	∞	0.0%	0.00%	fully suppressed
`haircolor`	Appearance & Body	7.9%	0.00%	∞	0.0%	0.00%	fully suppressed
`height`	Appearance & Body	7.5%	0.00%	∞	0.0%	0.00%	fully suppressed
`uniforms`	Attire, Role & Group	7.0%	0.00%	∞	0.0%	0.00%	fully suppressed
`ethnic_clothing`	Religion & Cultural	6.5%	0.00%	∞	0.0%	0.00%	fully suppressed
`nudity`	Appearance & Body	5.9%	0.00%	∞	0.0%	0.00%	fully suppressed
`troupe`	Attire, Role & Group	8.4%	0.00%	∞	0.0%	0.00%	fully suppressed
`weight`	Appearance & Body	8.8%	0.00%	∞	0.0%	0.00%	fully suppressed

Implication: The NETWORK channel is not a uniform filter — it is **attribute-selective**. Privacy defenses that target only the highest-risk retained/partial attributes (identity, gender, location, age, medical) will address >95% of actual NETWORK confirmed leakage. Fully-suppressed attributes (haircolor, height, weight, nudity, uniforms, formal, troupe, ethnic_clothing) need no special NETWORK-channel treatment, as the API boundary already blocks them completely.





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4. Cross-Group Synthesis

Q1: Does the stage gap hold inside every attribute family? Yes. For every family, raw-output leak rates exceed NETWORK confirmed rates by at least 6×. The gap is most extreme for Appearance/Body and Attire/Role attributes — all of which land in the "fully suppressed" regime — confirming these are consistently blocked at the API-call boundary regardless of what the model inferred.

Q2: Is text→text the dominant modality for NETWORK leakage? Yes and exclusively. text→text contributes 90.3% of all NETWORK-present items (3,495 of 3,870). image→text apps (e.g., snapdo, momentag) have near-zero NETWORK presence in the corpus — these apps process images locally and display results in-UI without external API calls. NETWORK leakage is therefore primarily a **text-input, cloud-API hazard.**

Q3: Does identity dominate NETWORK leakage in every app category? No — this is domain-specific. Identity dominates in Finance, Education, and Social categories (54–70% of NETWORK confirmed pairs). In Health apps, demographic attributes (`gender`, `age`) and `medical` lead instead. What is universal is that the "retained" regime attributes (those that break through the API boundary) are consistently the most sensitive semantic attributes regardless of category.

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5. Recommendations

1. **Prioritize identity redaction at API call boundaries.** Identity is the #1 confirmed attribute on NETWORK across all categories and is persist-dominant: it reaches the API because the app passes it through. Intercept API request bodies and redact name/person references before transmission.

2. **Apply output-side gender/age filters, not input-side.** Gender and age leak on NETWORK primarily through model inference (injection rate > persistence rate). Input-side filtering misses them; only post-generation output inspection catches them.

3. **Audit cloud-API apps first.** The binary presence structure means on-device apps pose zero NETWORK risk. Concentrate NETWORK-channel audits on the 14 apps that make external API calls.

4. **Focus on the top-5 high-NETWORK-leakage apps.** `waico` (3.21%), `nutri-track` (2.07%), `chat-driven-expense-tracker` (1.67%), `healyks` (1.40%), and `sgpa` (1.39%) account for the majority of NETWORK confirmed leakage. Targeted mitigations for these apps yield the highest risk reduction.

5. **Do not rely on NETWORK-channel filtering as a substitute for model-level intervention.** The 7.9×+ compression from raw output to NETWORK already occurs without explicit privacy engineering — but the residual 0.79% confirmed rate on NETWORK is attributable to the most privacy-sensitive attributes (identity, location, health). The tail is the threat.

6. **Exploit the fully-suppressed list as a safe baseline.** Eight attributes (`formal`, `haircolor`, `height`, `uniforms`, `ethnic_clothing`, `nudity`, `troupe`, `weight`) reach zero confirmed NETWORK leakage across all apps. Confirm this holds after model or app updates before removing monitoring for them.

Appendix

A. Setup and data quality

- Configs kept: 40 of 62 total
- Per-config sample cap: 200 items
- NETWORK presence: 60.3% of all attr-item pairs (channel structurally absent for apps with no external API calls)
- All rates conditioned on NETWORK presence unless stated otherwise

B. Figures generated

Figure	Description
`net_fig1_stage_gap`	3-stage gap (raw → any-ext → NETWORK) by app category
`net_fig2_attr_ranking`	NETWORK attribute ranking by any-leak and confirmed rate
`net_fig3_persist_inject`	Persistence vs Injection scatter per attribute (input→NETWORK)
`net_fig4_app_ranking`	Per-app NETWORK confirmed rate (apps with NETWORK present)
`net_fig5_channel_compare`	All-channel comparison (conditioned on presence)
`net_fig6_suppression_scatter`	Retention vs Suppression ratio scatter (raw-output→NETWORK)
`net_fig7_raw_vs_network`	Raw-output confirmed vs NETWORK confirmed per attribute
`net_fig8_category_family_pies`	Inferred attribute family breakdown per app category (NETWORK confirmed)